

the most sophisticated and high-end smartphone or best electric car) is the invention of super-efficient and long-lasting batteries. Systems known as uninterrupted power supplies will also serve the purpose of safeguarding vital constituents of the personal network in case of a power loss.

Endpoint

One of the most important measures of technology advancement is how much easier and simpler it makes life for us. Dream machines of the future, that is, PCs based on cutting-edge technology, will ensure that your refrigerator orders the orange juice when it is about to finish. You will not need to waste time standing in queues at supermarket cash counters—radio frequency identification tags in each of the items will automatically debit the required amount from your account.

Your car service centre will be able to detect car problems remotely using diagnostic software, and rectify these by downloading repair software straight on to your car using the Internet. Perhaps, most of your communication and interaction with appliances will be via voice using speech recognition and, moreover, digital assistants will be available to provide you with useful guidance.

Deep computing, a technology that emphasises on analysing data patterns and extracting valuable information from very large databases, is likely to become quite popular, and will be accessible by even small enterprises. With its origin in IBM chess-playing computer Deep Blue,

deep-computing techniques will be used in various sectors such as simulation of new medicines and chemical processes.



Fig. 6: RFID tags in each of the purchased items will automatically debit the required amount (Image courtesy: <https://adcsolutions.wordpress.com>)



Fig. 7: Deep-computing techniques will be used in various sectors, for example, in simulations of new medicines (Image courtesy: www.atelier.net)



Fig. 8: In future, our PC network could very well be growing in our brain itself (Image courtesy: www.futuretimeline.net)

Drugs will no longer be designed by mixing chemicals and testing on animals, as scientists will do the efficacy testing on a computer. Personal computers will be able to recognise the current user and visually present customised interfaces. For example, interface for a young child will be different from an interface for a parent.

There are no physical limits to improvements and progress in computer technology. Various aspects such as bandwidth, storage and compute power are likely to evolve, and you can expect to see several manifold improvement. One of the main reasons for this optimism is that the amount of power required in a compute step is not as much as thought earlier.

The quantum of energy needed to execute one compute operation is close to the energy of the atom itself, and it is conceivable to store several bits in an atom. In times to come, personal computing will evolve into intra-personal and intra-cellular beings. There will be a time when, perhaps, each human neuron will be commandeered by a molecular network that grows and repairs itself.

Computers will become networks of polymer filaments growing inside and together with a human being. Probably, elements of these polymer filament based networks will be inserted into embryos in the first month itself of their growth. These will grow to develop into an enormous network inside the brain.

Hence, computers will be inside each of us from birth, and thereafter these will unite all living beings into an integrated computing network. **EFY**